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Mini Project Synopsis

on

"AI BASED DRIVER DROWSINESS DETECTION SYSTEM"

By

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1. Aim:

The aim of the **AI-Based Driver Drowsiness Detection System** is to **develop a real-time monitoring system that detects signs of driver fatigue using artificial intelligence and computer vision techniques**. The ultimate goal is to **enhance road safety by reducing fatigue-related traffic incidents** using a cost-effective and easily deployable solution.

2. Introduction:

Driver fatigue is a major cause of road accidents worldwide, leading to significant loss of life and property. Many accidents occur due to drivers becoming inattentive or falling asleep while driving. Traditional methods to detect drowsiness, such as wearable sensors or vehicle-mounted devices, are often expensive, intrusive, and not feasible for everyday use.

The **AI-Based Driver Drowsiness Detection System** provides a non-intrusive and cost-effective solution by leveraging computer vision and artificial intelligence techniques. The system uses a standard webcam to monitor the driver's facial features in real time, analysing eye closure, blinking patterns, and yawning to detect signs of fatigue. Upon detecting drowsiness, it immediately alerts the driver through visual and audio signals, thereby enhancing road safety and preventing accidents caused by inattentive driving.

3. Motivation

Road accidents caused by driver fatigue are a serious concern worldwide, resulting in loss of life and property every year. Existing solutions, such as wearable sensors or advanced vehicle-mounted systems, are often expensive, intrusive, and not easily accessible for everyday drivers.

This project was chosen to **develop a simple, cost-effective, and non-intrusive solution** using AI and computer vision techniques. It also provides an opportunity to gain hands-on experience in machine learning, computer vision, and real-time system development.

4. Problem Statement

Driver fatigue is one of the leading causes of road accidents globally, resulting in significant loss of life and property each year. Many accidents occur because drivers become inattentive or fall asleep while driving, especially during long journeys or night shifts. Existing solutions, such as wearable sensors or vehicle-mounted monitoring devices, are often expensive, intrusive, and not practical for widespread everyday use.

Therefore, there is a need for a **cost-effective, non-intrusive, and real-time monitoring system** that can detect signs of driver drowsiness, such as prolonged eye closure, frequent blinking, or yawning, and immediately alert the driver to prevent accidents. This project aims to address this problem by developing an AI-based system using a standard webcam and computer vision techniques to enhance road safety.

5. Proposed Solution

The proposed system uses a **computer vision and AI-based approach** for real-time driver drowsiness detection. The system captures video frames from a standard webcam, detects the driver's face and eyes using **Mediapipe facial landmark detection**, and then classifies the eye state (open or closed) using a **Convolutional Neural Network (CNN)** trained on labelled eye images. By continuously monitoring eye closure duration and blink patterns, the system can detect signs of drowsiness and immediately trigger visual and audio alerts to warn the driver.

Compared to existing systems that rely on expensive wearable sensors or vehicle-mounted devices, this approach is **non-intrusive, cost-effective, and easily deployable**. It provides real-time monitoring without requiring additional hardware, works with a standard laptop or webcam, and leverages AI to improve detection accuracy. Furthermore, the system can be easily extended to monitor multiple drivers or integrated with mobile and IoT devices for wider applications in road safety.

6. Expected Output

The main objectives of the **AI-Based Driver Drowsiness Detection System** are:

- **Real-time Monitoring:** To continuously monitor the driver's facial features, such as eye closure and blinking patterns, to detect signs of drowsiness.
- **Alert Generation:** To provide immediate visual and audio alerts when driver fatigue is detected, preventing accidents caused by inattention.
- **Cost-Effective and Non-Intrusive:** To implement a system that uses a standard webcam and AI, eliminating the need for expensive wearable sensors or vehicle-mounted devices.
- **High Accuracy Detection:** To achieve reliable and accurate drowsiness detection using AI and computer vision techniques, even under varied lighting conditions and head movements.
- **Data Logging:** To maintain a log of drowsiness events for later analysis and improvement of detection thresholds.
- **Scalability:** To design a system that can be extended for mobile or IoT-based deployment in vehicles.

7. References

Paper-1: Base Paper

Sengar, S. S., et al. (2024). Artificial Intelligence-based Real-time Driver Drowsiness Detection. arXiv.

Paper-2: Reference Paper – 1

Albadawi, Y., Takruri, M., & Awad, M. (2022). A Review of Recent Developments in Driver Drowsiness Detection Systems. Sensors, 22(5), 2069.

Paper-3: Reference Paper – 2

Rahman, A. (2022). Computer Vision-based Approach to Detect Fatigue Driving. ScienceDirect.

Paper-4: Reference Paper – 3

Hassan, O. F., et al. (2025). Real-time Driver Drowsiness Detection Using Transformer Architectures: A Novel Deep Learning Approach. Scientific Reports.

Paper-5: Reference Paper – 4

Bhanja, A., et al. (2025). A Real-time Driver Drowsiness Detection System. ROBOMECH Journal.